

AI Accelerators

#AI

Companies: Nvidia, AMD, Avago, MediaTek, TSMC, ASE, Amkor

Stefan Chang

+852 9250 7088

stefan.chang@aletheia-capital.com

Warren Lau

+852 9181 4766

warren.lau@aletheia-capital.com

An Early Look at 2027 Accelerators

We are launching our 2027E GPUs and ASICs estimates. In this note, we will cover 1) production volume estimates; 2) CoWoS allocation; and 3) HBM consumption. From a production angle, we estimate NVDA will continue to take the lead with mid-40% unit share and more than half of CoWoS consumption. TPU volume will fall behind NVDA's GPU volume by only a small margin. In terms of CoWoS consumption, however, we expect AMD to grow the most and to consume more than TPU CoWoS wafer to support its Venice CPU. On the flip side, AWS will have less than sufficient CoWoS support for its Trainium 3 and will need EMIB-T packaging from Intel to make up the volume. Lastly, total HBM density demand from accelerators will reach 52b-Gb in 2027E (+77% YoY) based on our GPUs and ASICs production estimates, despite recent development of HBM D/G by GPUs and TPUs. Incremental server DRAM density is <8b-Gb if all saved HBM capacity is converted to server DRAM and is insignificant vs the current server DRAM demand/supply gap.

21m GPU+ASICs production in 2027E, up from 13.4m in 2026E

We estimate total GPUs and ASICs production volume from the top four (NVDA, AMD, Google's TPUs and AWS's Trainium) to reach 21m in 2027E, up from 13.4m this year. Production volume for NVDA's GPUs could reach 9.0-9.5m in 2027E and is still the highest among all, followed by TPUs at 8.0-8.5m (with upside potential should there be enough capacity). AVGO will still be the key supplier for TPUs in 2027E with 50%+ unit share and 80%+ value share. AMD could also produce up to 1.6m GPUs; its top three customers (OAI, Meta and Anthropic) will consume 70-80% of these. Lastly, we estimate AWS's Trainium 3 volume to be only 2m because of CoWoS capacity constraint from foundry and OSAT partners. AWS will need EMIB-T packaging from Intel to make up the volume.

CoWoS allocation: NVDA still the largest; AMD to grow the most

We estimate TSMC's annual CoWoS output to grow to 2.2m in 2027E, almost double YoY. Including OSATs, total CoWoS supply in 2027E could reach 2.6-2.7m, up from ~1.2m this year. NVDA will unsurprisingly consume the most CoWoS wafer with 1.2m/1.4m at TSMC/total, still 50%+ of the total market. AMD will be the second largest CoWoS wafer buyer in the market to support both its Instinct GPUs and Venice CPUs, even higher than the total CoWoS wafers for TPUs by AVGO and MTK combined (450-500k). Lastly, we estimate AWS can only get <100k CoWoS wafer support from TSMC and OSATs, which may explain why it is aggressively working with Intel for its Trainium 3 chips.

No impact to DRAM demand/supply from HBM D/G

Based on our accelerator production volume estimates, total HBM consumption will be 52b-Gb in 2027E, +77% YoY from 29b-Gb in 2026E. Considering inventory and production lead-time, HBM production should be 30-35b-Gbs and 55-60b-Gbs in 2026/27E, which is about the same as the current production plans by the memory makers. Notably, we have incorporated the recent development of HBM de-spec by GPU and ASIC makers.

We believe NVDA is working on HBM4E 8H (32GB per stack, 256 GB total), HBM4 12H (288GB), and HBM4 8H (192GB) for its Rubin Ultra (RU), in addition to the original plan of HBM4E 12H (384 GB). It is still too early to conclude which spec NVDA will adopt or if RU will have multiple SKUs. We think HBM4E 8H is the most likely solution. Despite 11% density downgrade from Rubin (288GB), HBM4E provides at least 20-30% wider memory bandwidth. NVDA may also enhance its performance by some optimization or external memory/storage (continued in the next page).

Further, NVDA is not alone. We are aware that AVGO's TPUv8.5 (Whalefish), which consists of 2x Sunfish (TPUv8i) CoW module on a single giant substrate, also plans to downgrade its HBM adoption from HBM3E 12H (for Sunfish) to HBM3E 8H. HBM density for Whalefish (24GB per stack, 16x slot, total 384GB) will still be higher than that for Sunfish (36GB per stack, 8x slot, total 288GB), but density per stack will drop to 24GB from 36GB. Lastly, we still assume AMD's MI500X will be equipped with HBM4E 12H, and HBM density per GPU will reach 384GB (2x base-die version). This will continue AMD's HBM density advantage vs NVDA's RU at 1.5:1 ratio, the same as MI455x vs Rubin.

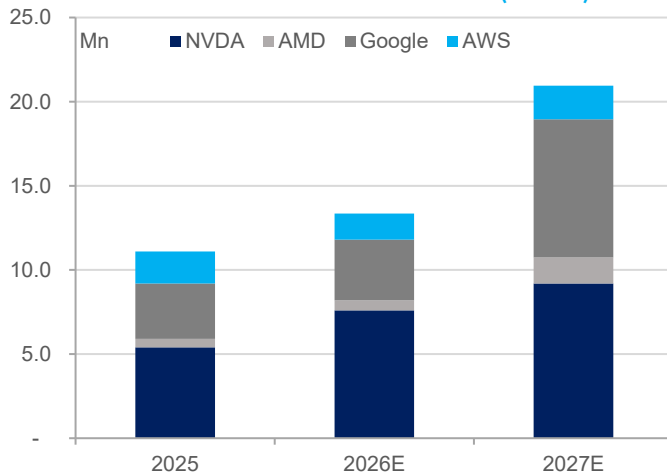
To conclude, with the de-spec discussed above, we estimate total HBM density saving is <2.5b-Gb from Rubin Ultra and Whalefish D/G. This means <8b-Gb server DRAM using a 1-3 trade ratio (form HBM vs server DRAM) even if the saved wafer capacity from HBM de-spec is 100% converted to server DRAM. This is insignificant compared to the current server DRAM demand/supply gap, which could be as big as 100b-Gb. As such, we do not see any impact to the DRAM demand/supply dynamics.

FIGURE 1: CHIP UNIT SHIPMENT BY CHIPMAKERS

Chip Volume (m)	2025	2026E	2027E
NVDA	5.4	7.6	9.2
Hopper	0.3	-	-
B200/B300	5.1	5.4	0.7
Rubin	-	2.2	6.5
Rubin Ultra	-	-	2.0
AMD	0.5	0.6	1.6
MI355X	0.5	0.4	-
MI450X/455X	-	0.3	1.5
MI550X	-	-	0.1
Google	3.3	3.6	8.2
TPUv5P (Viperfish)	0.8	-	-
TPUv6E (Ghostlite)	2.0	-	-
TPUv7 (Ghostfish)	0.5	2.5	-
TPUv8i (Sunfish)	-	0.5	4.0
TPUv8t (Zebrafish)	-	0.6	4.0
TPUv8i+ (Whalefish)	-	-	0.2
AWS	1.9	1.6	2.0
Trainium 2.0/2.5	1.9	0.1	-
Trainium 3	-	1.5	2.0
Total	11.1	13.4	21.0

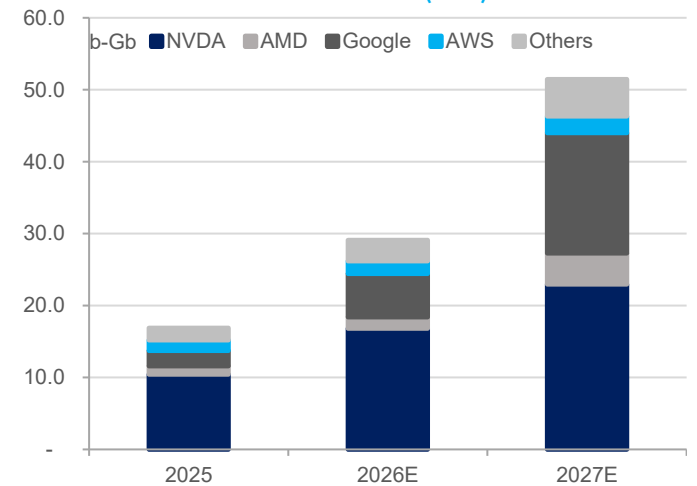
Source: Aletheia/TAG

FIGURE 2: CHIP UNIT SHIPMENT BY CHIPMAKERS (m units)



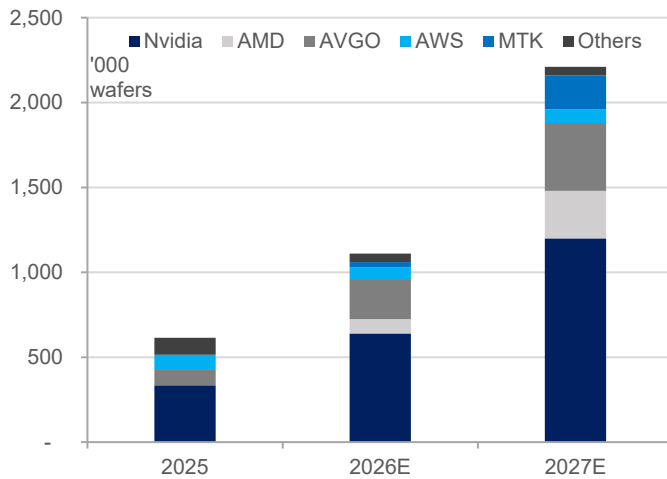
Source: Aletheia/TAG

FIGURE 3: HBM DEMAND BY CHIPMAKERS (b-Gb)



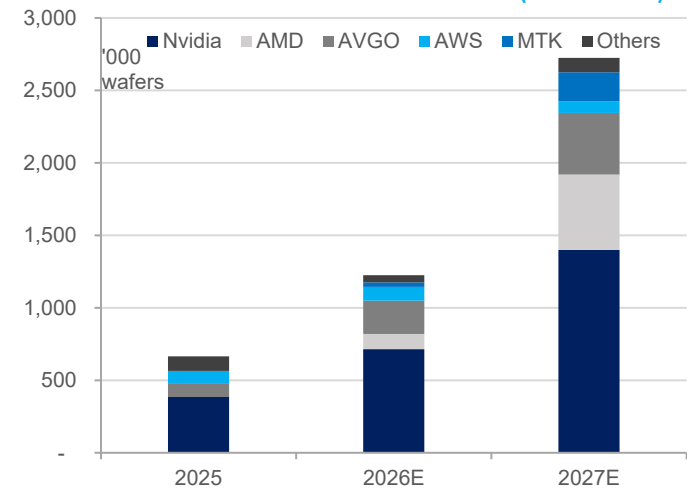
Source: Aletheia/TAG

FIGURE 4: TSMC'S COWOS CONSUMPTION BY CHIPMAKERS



Source: Aletheia/TAG

FIGURE 5: COWOS CONSUMPTION BY CHIPMAKERS (TSMC + OSAT)



Source: Aletheia/TAG

Alētheia Research Team

Macro/Strategy	LEAD ANALYST	Sector	LEAD ANALYST
Global Strategy	 Jonathan Wilmot	Technology Hardware	 Warren Lau
Asianomics	 Dr. Jim Walker	Consumer & Internet	 Nirgunan Tiruchelvam
Tech Thematic Strategy	 Keith Woolcock	China Technology	 Eric Wen
Asia Equity Strategy	 David Scott	Telecoms	 Chris Hoare
China Strategy	 Vincent Chan	CrossASEAN	 Angus Mackintosh
Tactical Alpha Strategy	 Justin Collazo	India	 Maulik Patel
Global Commodities	 Steven Schlegel	MENA	 Jaap Meijer

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● Al Park 1 201 942 0539 al.park@aletheia-capital.com AMERICAS - NEW YORK	● Linda Gustafsson 44 791 91 21 365 linda@nordlinkcapital.com EUROPE - NORDICS	● Michael Chambers 65 9858 9759 michael.chambers@aletheia-capital.com ASIA - SINGAPORE	● Graeme Bateman 852 2534 7437 graeme.bateman@aletheia-capital.com ASIA - HONG KONG
● Wayne Chang 1 646 256 3366 wayne.chang@aletheia-capital.com AMERICAS - NEW YORK	● Daren Riley 44 748 31 59 865 daren.riley@aletheia-capital.com EUROPE - LONDON	● Richard Wallace 852 2534 7429 richard.wallace@aletheia-capital.com ASIA / EUROPE - HONG KONG	
● Terrence Alford 1 469 403 9936 terrence.alford@aletheia-capital.com AMERICAS - DALLAS	● Dhananjay Mahurkar 44 777 55 26 870 dhananjay.mahurkar@aletheia-capital.com EUROPE - LONDON	● Augustine Chen 8869 2739 8793 augustine.chen@aletheia-capital.com ASIA - HONG KONG	

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